

Ideation

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POTENTIAL PROJECT IDEAS

The team generated six candidate applications, each aimed at a problem a member had encountered directly on campus. All six were assessed for feasibility within one semester, for the availability of data we could legitimately obtain, and for the amount of substantive web engineering work they would require.

Financial Buddy A student-focused financial planning assistant for budgeting income, expenses, savings, and investment education.

Student Information Assistant A conversational university assistant that helps students navigate campus resources, policies, deadlines, and departments.

Campus Events & Community Hub A centralized platform for discovering campus organizations, events, and activities based on student interests.

Student Part-Time Job Matcher A platform connecting students with campus jobs and surfacing opportunities based on skills, schedules, and interests.

Campus Marketplace A verified student-to-student marketplace for textbooks, furniture, electronics, and other campus needs.

QuadConnect (selected) A verified campus social discovery platform that helps students form friendships and communities through structured, in-person experiences.

IDEA SELECTION

The team selected **QuadConnect** because it addresses a concrete student problem while offering a clear opportunity for data-driven personalization and interactive web design. Unlike conventional social platforms that emphasize profiles, swiping, and prolonged messaging, QuadConnect is designed around the transition from online discovery to an actual shared experience. The concept also provides a strong foundation for future implementation through matching algorithms, preference-based recommendations, scheduling, check-in workflows, and feedback-driven improvement.

The remaining five ideas were set aside for specific reasons. Financial Buddy and the Student Information Assistant both depend on data the team cannot obtain reliably, namely personal banking records and authoritative university policy text. The Events Hub and the Marketplace reduce largely to listing-and-search applications, which limits their algorithmic depth. The Job Matcher was the closest runner-up, but its value depends on employer participation the team cannot guarantee within a single semester.

PROBLEM SPACE EXPLORED

Students have many opportunities to meet people at a university, but discovering compatible peers and turning that discovery into a comfortable, structured interaction can be difficult. Relevant information is often fragmented across social media, student organizations, and event pages. Existing platforms also tend to require users to initiate conversations without providing a natural activity or context for the interaction.

DESIGN DIRECTION

QuadConnect focuses on **verified students, compatibility, and real-world experiences**. Students authenticate through university SSO, describe their interests and social preferences, select a connection type, and provide availability. The system can then identify compatible participants and organize a weekly experience at an approved campus location. The current prototype focuses on **Friend Connect** and **Squad Connect**, with other connection types suitable for future expansion.

CORE DESIGN PRINCIPLES

- **Verified community:** restrict participation to university students through institutional authentication.
- **Intentional matching:** use interests, social preferences, availability, and campus involvement rather than random pairing.
- **Offline-first interaction:** give students a reason and a structure to meet in person.
- **User control and safety:** use approved public locations, check-in, reporting, and private feedback.
- **Feedback loop:** use post-experience feedback to improve future matching and experiences.

PROJECT ARTIFACT

The accompanying wireframes visualize the proposed user flow and interface. Interactive prototype: <https://trek-galaxy-20520998.figma.site/>

QuadConnect: A Verified Campus Social Discovery Platform

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Abstract—Team: The Connectors | Team Number: 13. QuadConnect is a verified campus social discovery platform designed to help university students form meaningful relationships through structured, in-person experiences. Students authenticate through university SSO and provide information about interests, campus involvement, social preferences, availability, and preferred activities. The system uses these inputs to identify compatible students for one-to-one Friend Connect or small-group Squad Connect experiences. Rather than centering the experience on endless messaging, QuadConnect provides a scheduled meeting at an approved campus location, suggested activities, check-in, and private post-experience feedback. The prototype demonstrates the complete flow from onboarding and preference collection through matching, meeting, and feedback.

I. INTRODUCTION

University campuses contain large and diverse student communities, yet students can still have difficulty forming new social connections, especially when they are new to campus or outside established communities. Social discovery is also fragmented across separate services for student organizations, events, messaging, and social networking. QuadConnect explores a different model in which technology is used not simply to facilitate online interaction, but to create a clear path toward a real-world experience.

II. MOTIVATION / WHY THIS IDEA

The key problem is not a lack of people to meet; it is the difficulty of finding compatible people and a comfortable context in which to meet them. Starting a conversation with an unfamiliar student can feel awkward when there is no shared activity or purpose. QuadConnect addresses this friction by combining compatibility-based discovery with a scheduled activity in a public campus setting.

The concept is interesting from a computing perspective because matching is not based on a single attribute: the system must combine hard constraints such as availability and connection type with softer signals such as interests, social energy, preferred environments, campus involvement, and activity preferences. Modelling those signals so that they are both queryable and explainable is the central technical challenge.

*All team members are considered equal contributors. In special cases, like mentioned in "Authorship and Acknowledgement," the naming will follow order of contribution / ownership ratio.

III. PROPOSED FEATURES

- **UIUC SSO verification:** establish a verified student-only community.
- **Interest and campus profile:** capture hobbies, interests, RSOs, college/department, and optional work information.
- **Connection selection:** choose Friend Connect or Squad Connect in the current prototype.
- **Social preferences:** capture social energy, conversation style, preferred environments, group size, age and degree-level preferences, and community preferences.
- **Availability and activities:** collect time windows and preferred activities such as coffee, food, sports, campus walks, games, and study or social spaces.
- **Compatibility matching:** identify participants who satisfy required constraints and have strong overall compatibility.
- **Scheduled experience:** provide a date, time, approved campus location, and suggested activity.
- **Check-in and feedback:** support attendance confirmation and check-in, plus private post-experience ratings and connection preferences.

IV. APPLICATION FUNCTIONALITY

A student begins by signing in through UIUC SSO and completing a short onboarding flow. They select interests and campus involvement, choose a connection type, define social preferences, and provide weekend availability. After enrollment in a weekly matching cycle, QuadConnect evaluates eligible participants. For Friend Connect, the system identifies a compatible one-to-one connection; for Squad Connect, it forms a small group around shared interests, availability, and compatible social preferences.

The resulting match displays why participants were paired and provides a suggested experience at an approved campus location. Before the meeting, participants confirm attendance and receive check-in instructions. Afterward, they submit private feedback and can indicate which participants they would like to stay connected with; a lasting connection is created only when interest is mutual. Future versions can add connection types, campus events, student opportunities, and an AI-powered campus assistant.

NOTE ON ACKNOWLEDGEMENT AND AUTHORSHIP

Not applicable: the concept originates in class brainstorming rather than prior coursework or an existing project, and all members are equal contributors.

REFERENCES

- [1] *Tools, inspirations, and related work consulted while developing the concept and its interface design.*
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